

CPSC 375

Introduction to Data Science and Big Data Analytics

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What we will cover today

- Exploratory Data Analysis
 - In Python

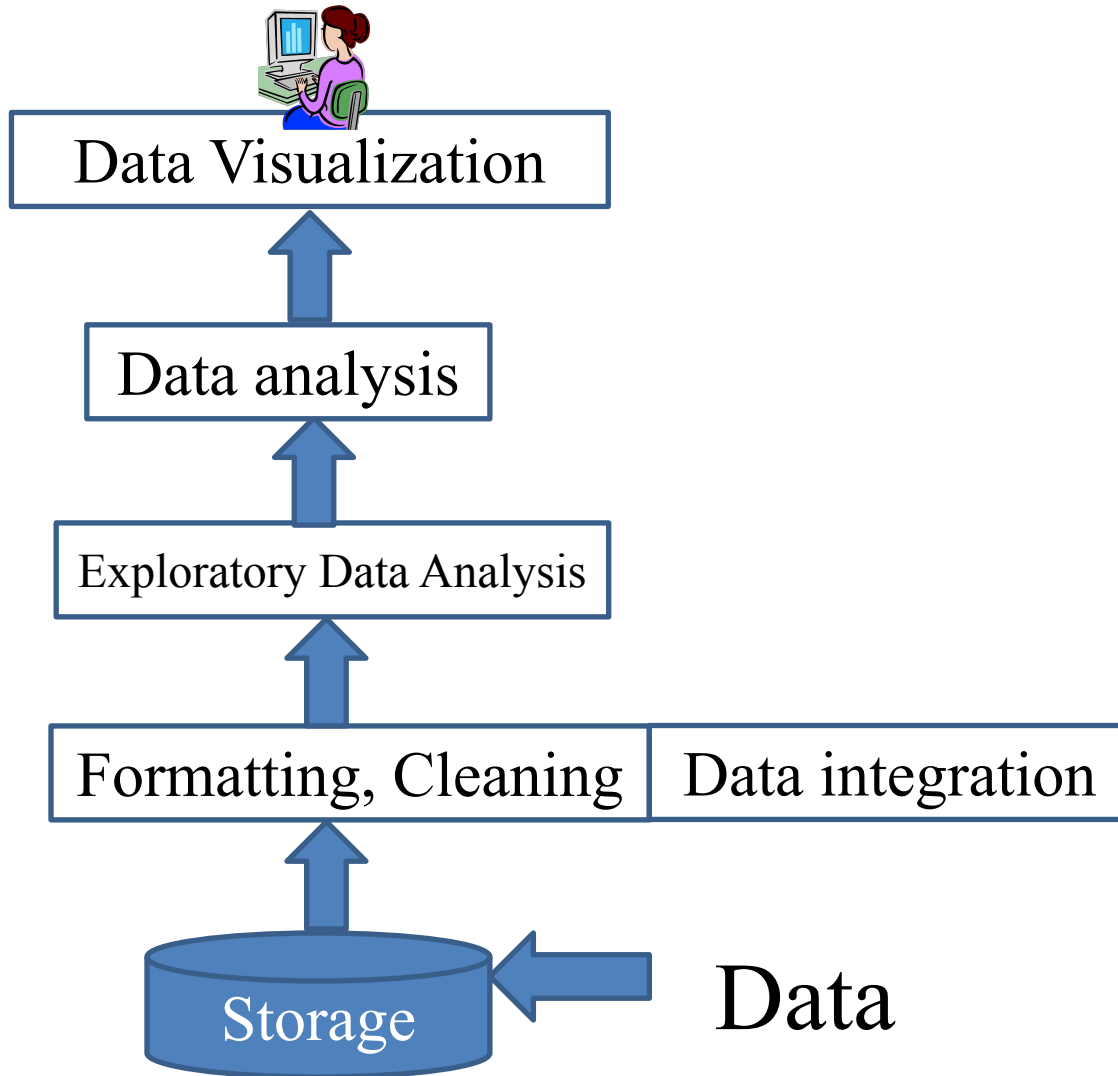
Exploratory Data Analysis (EDA)

- First step to building a model
- Data-driven (model-free)
- Goal: Get a general sense of the data
 - Getting your hands “dirty” with the data
- Interactive and visual
 - Humans are good at pattern recognition

Why EDA?

- To gain intuition about the data
- Sanity checking
 - Is the data as expected?
 - Are the ranges of variables as expected?
- detect outliers (e.g. assess data quality)
- test assumptions (e.g. normal distributions or skewed?)
- identify useful raw data & transforms (e.g. $\log(x)$)
- Is there missing data?
- What is the scale of the data?
 - Is it Big Data?

The Data Science Process



Exploratory Data Analysis (EDA)

- Systematically going through the data
 - Generating summary statistics
 - Evaluating data quality
 - Visualizing each variable
 - Visualizing all pairwise relationships

Summary Statistics

- *not* visual
- sample statistics of data X
 - mean
 - median
 - quartiles of sorted X : Q1 value = $X_{0.25n}$, Q3 value = $X_{0.75n}$
 - interquartile range: $\text{value}(Q3) - \text{value}(Q1)$
 - `IQR()`
 - Variance
 - mode: most common value in X
- `iris.describe()` or `iris.describe().T`

Data quality

- Is there missing data?
 - Count the NAs
 - Possible that missing values are not stored as NA
 - Instead, 0 or -1 or something that cannot be a valid value
- Are there outliers?
- Check for each variable

Numerical vs Categorical Variables

- Numerical variables
 - Values can be **ordered**, i.e., there is a natural ordering between the values
 - E.g., Age (younger/old), Weight (lighter/heavier)
 - Can calculate averages
 - Also called **Ordinal** variables
- Categorical variables
 - There is no natural ordering between the values
 - E.g., Marital status: *Single, Married, Divorced*
 - E.g., Color: *red, blue, green, ...*
 - E.g., Species: *iris.Species: setosa, versicolor, virginica*
 - Cannot calculate averages
 - Also called **Nominal** variables

Some special values

- `np.NaN`, `pd.NaT`, `pd.NA`: Not Available (placeholder for a missing value).
- `None`: empty value
- `np.Inf`: Infinity
- `pd.isna()` and `pd.isinf()`
 - to identify missing, empty and infinite values in data
- `pd.notna().all(axis='columns')` to identify the rows in a data frame that are complete (no missing values)
- `df.dropna()` return a data frame with incomplete rows removed
- Many NumPy functions give NA if *any* of the inputs is NA
 - E.g.: `np.mean(x)`
 - Use `nan*` version to ignore the NAs in the input, e.g.: `np.nanmean(x)`
- Pandas functions ignore rows if *any* of the inputs is NA

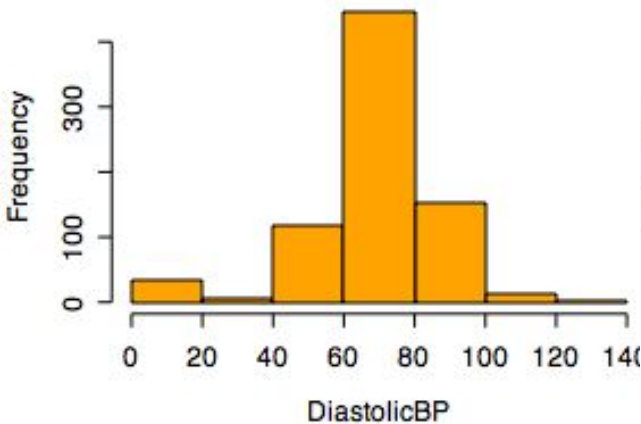
Single Variable Visualization

- Numerical variable:
 - Histogram
 - Box plot
- Categorical variable:
 - Bar graph

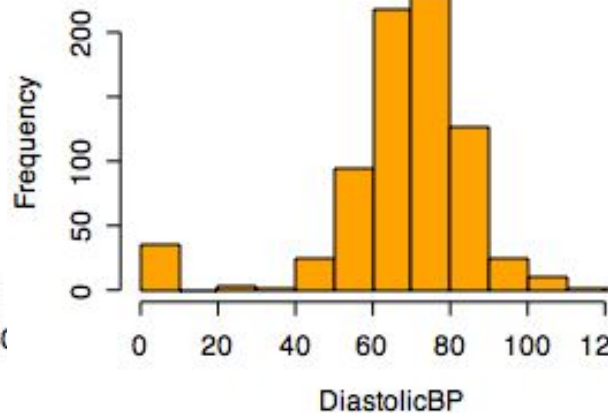
Single Variable Visualization

- Histogram:
 - Shows center, variability, skewness, modality, outliers, or strange patterns.
 - Bin width and position matter
 - Beware of real zeros

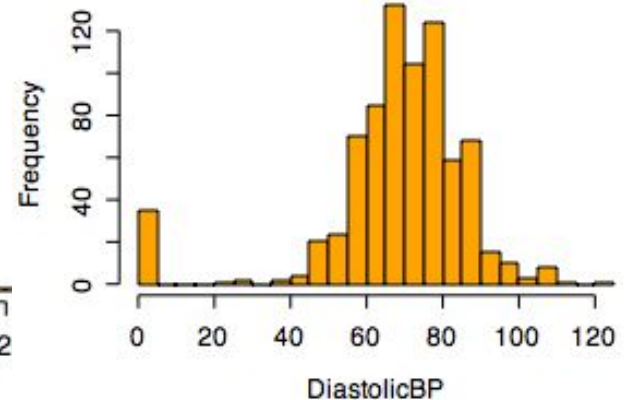
Histogram of DiastolicBP



Histogram of DiastolicBP



Histogram of DiastolicBP



Python code

- `ggplot(data=airquality)+geom_histogram(mapping = aes(x='Ozone'))`
- `airquality.Ozone.hist(bins=30)`
- `import seaborn as sns`
`sns.histplot(data=airquality, x='Ozone', bins=30)`

Issues with Histograms

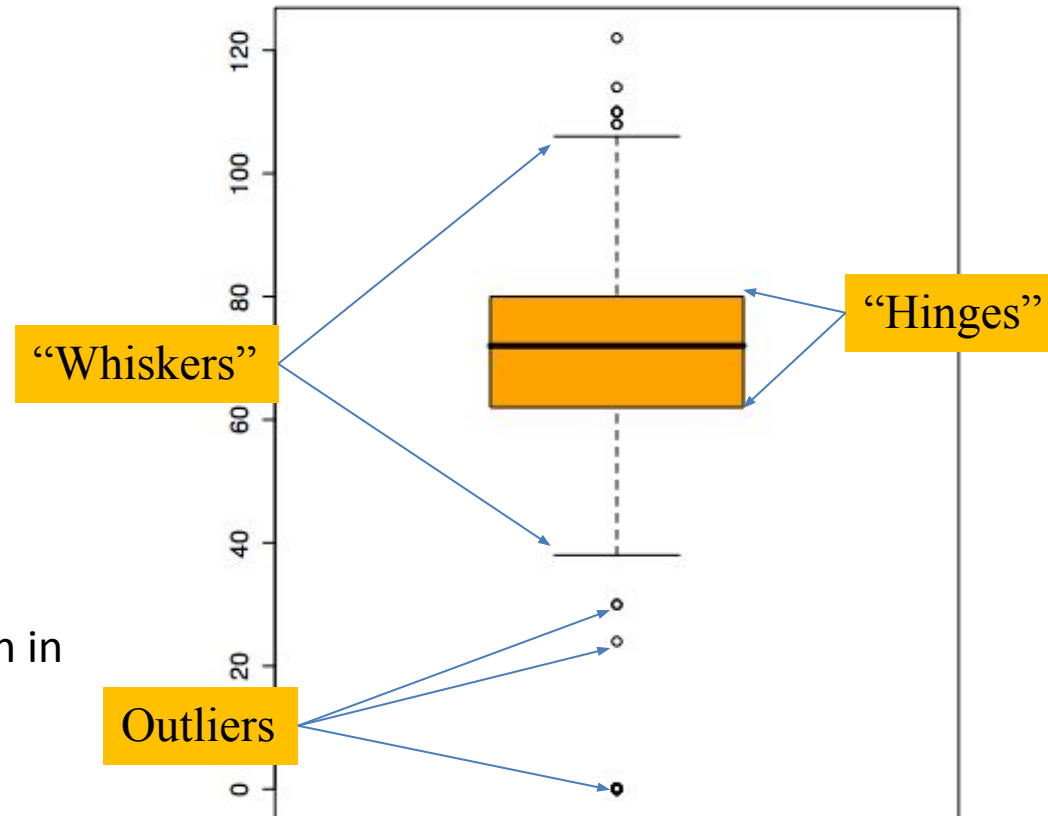
- For small data sets, histograms can be misleading
 - Small changes in the data, bins, or anchor can deceive
- **For large data sets, histograms can be quite effective at illustrating general properties of the distribution**
- Histograms only **work with one variable at a time**

Classwork

- [soodk6/cpsc375-classwork](#)
–eda.ipynb – Part 1

Boxplots

- Shows a lot of information about a variable in one plot
 - Median
 - Inter-quartile range (IQR)
 - Outliers
 - Range
 - Skewness
- Negatives
 - Overplotting
 - Hard to tell shape of distribution
 - no standard implementation in software (many options for whiskers, outliers)



The Five-Number Summary

- Median (50% value)
- Quartiles
 - 1st and 3rd quartiles: 25% and 75% values
- Extremes: maximum and minimum of the expected **valid** points

Statistic	Value
Max	8.67
Q3	2.93
Median	1.90
Q1	1.15
Min	0.20

Making Boxplots

- A **boxplot** is a graphical display of the **five-number summary**

Identifying Outliers

In addition to serving as a measure of spread, the interquartile range (*IQR*) is used as part of a rule of thumb for identifying outliers.

The 1.5 x *IQR* Rule for Outliers

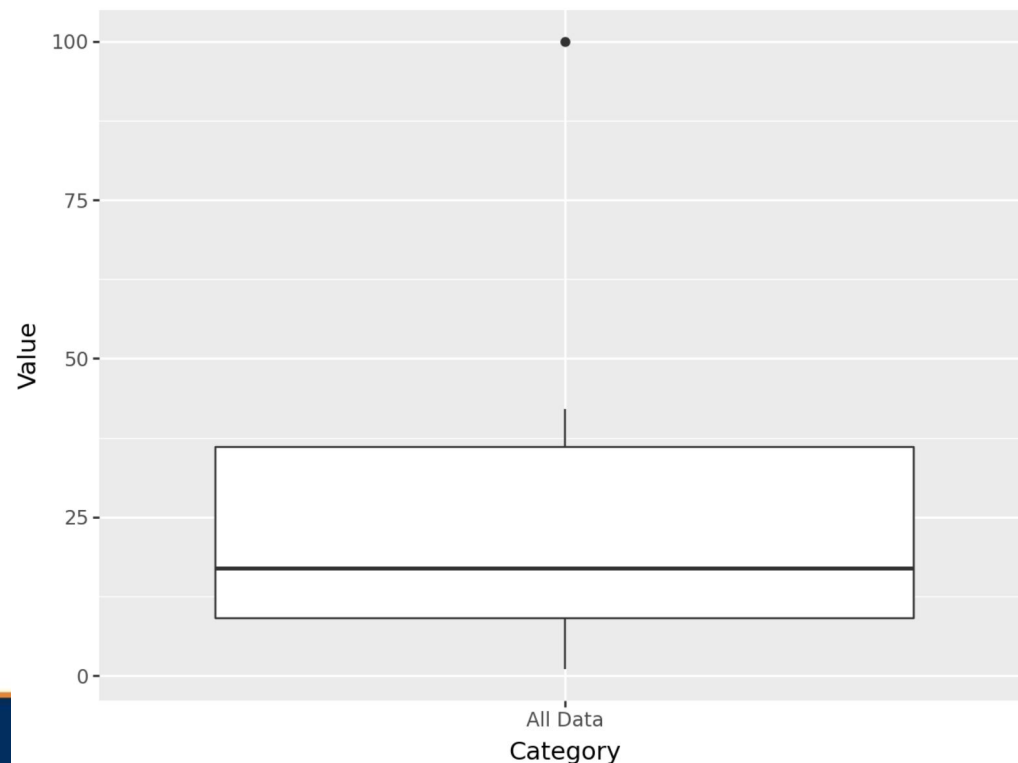
Call an observation an outlier if it falls more than 1.5 x *IQR* above the third quartile or below the first quartile.

Interquartile range (*IQR*) = 0.75 quantile – 0.25 quantile

Only a rule of thumb – not every identified data point is an outlier

Demonstration

- Construct a boxplot of 9 values:
36, 42, 100, 28, 17, 12, 9, 4, 1



Boxplot in Lets-Plot

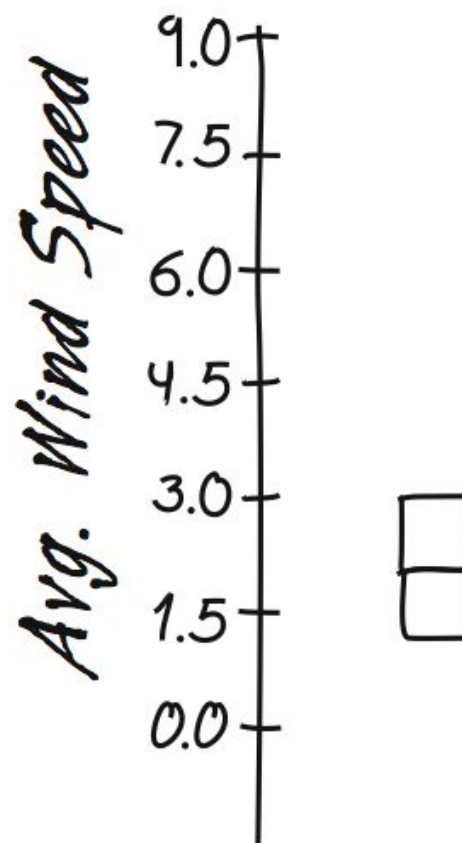
- `geom_boxplot()`
- Hinges are at the 25% and 75% quartiles
- The upper whisker extends from the hinge to the highest value that is within $1.5 * \text{IQR}$ of the hinge
- The lower whisker extends from the hinge to the lowest value within $1.5 * \text{IQR}$ of the hinge.
- Data beyond the end of the whiskers are outliers and plotted as points

Boxplots of iris variables

- `ggplot(data=iris) +
 geom_boxplot(mapping=aes(y='Sepal.Width'))`
- `iris.boxplot('Sepal.Width')`
- `sns.boxplot(iris, y='Sepal.Width')`

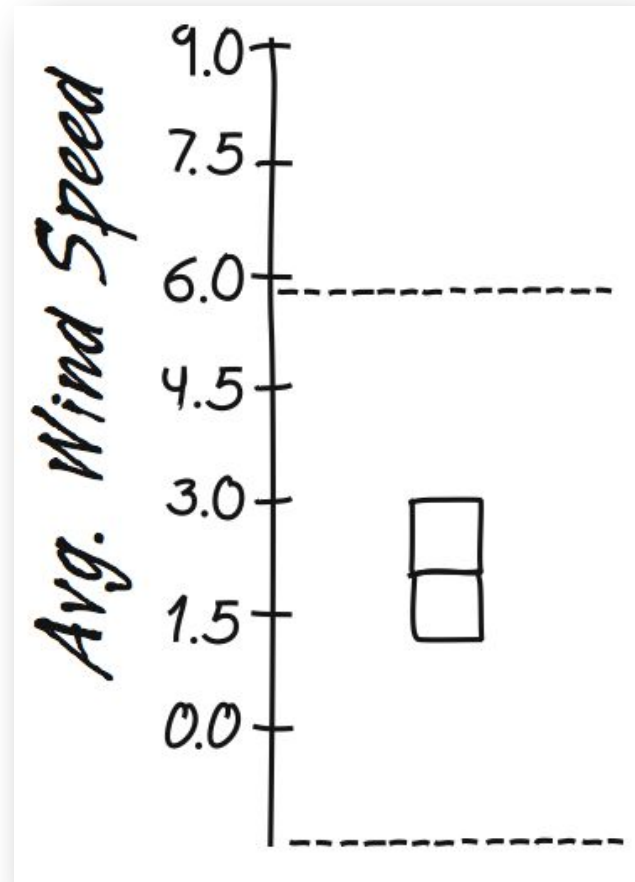
Constructing Boxplots

1. Draw a single vertical axis spanning the range of the data
2. Mark the median
3. Draw “hinges” at the lower and upper quartiles
4. Then connect them with vertical lines to form the “box”



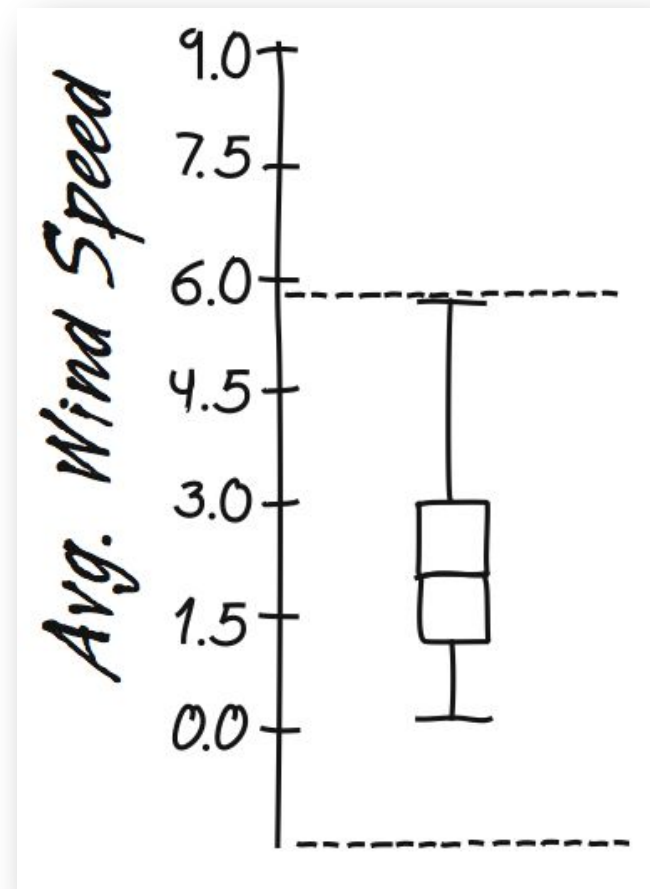
Constructing Boxplots (cont.)

2. Draw “fences” around the main part of the data
- The upper fence is $1.5 \times (\text{IQR})$ above the upper quartile.
 - The lower fence is $1.5 \times (\text{IQR})$ below the lower quartile.
 - Note: the fences only help with constructing the boxplot and should not appear in the final display.



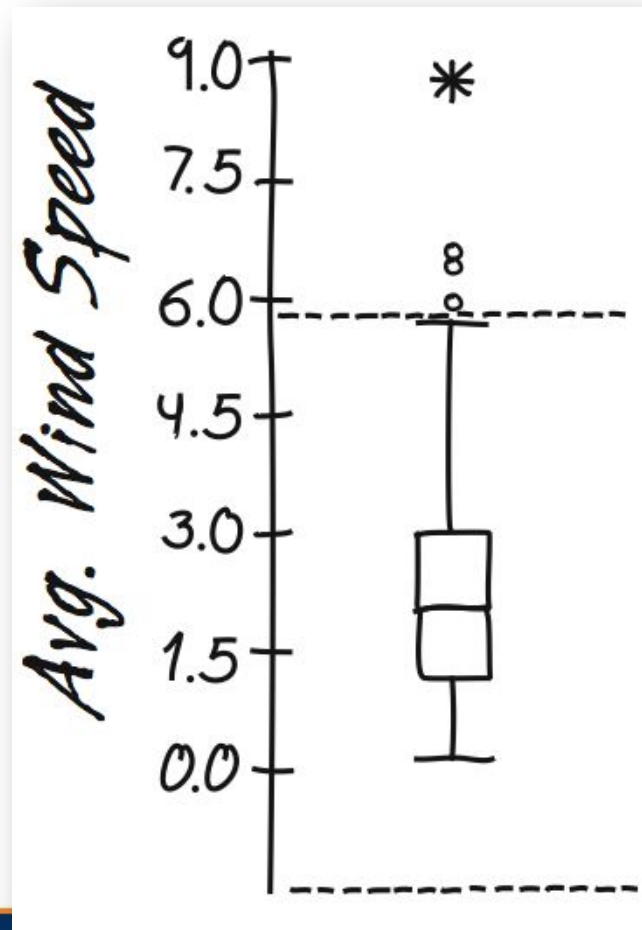
Constructing Boxplots (cont.)

3. Use the fences to grow “whiskers.”
- Draw lines from the ends of the box up and down to the *most extreme data values found within the fences*.
 - If a data value falls outside one of the fences, we do *not* connect it with a whisker.

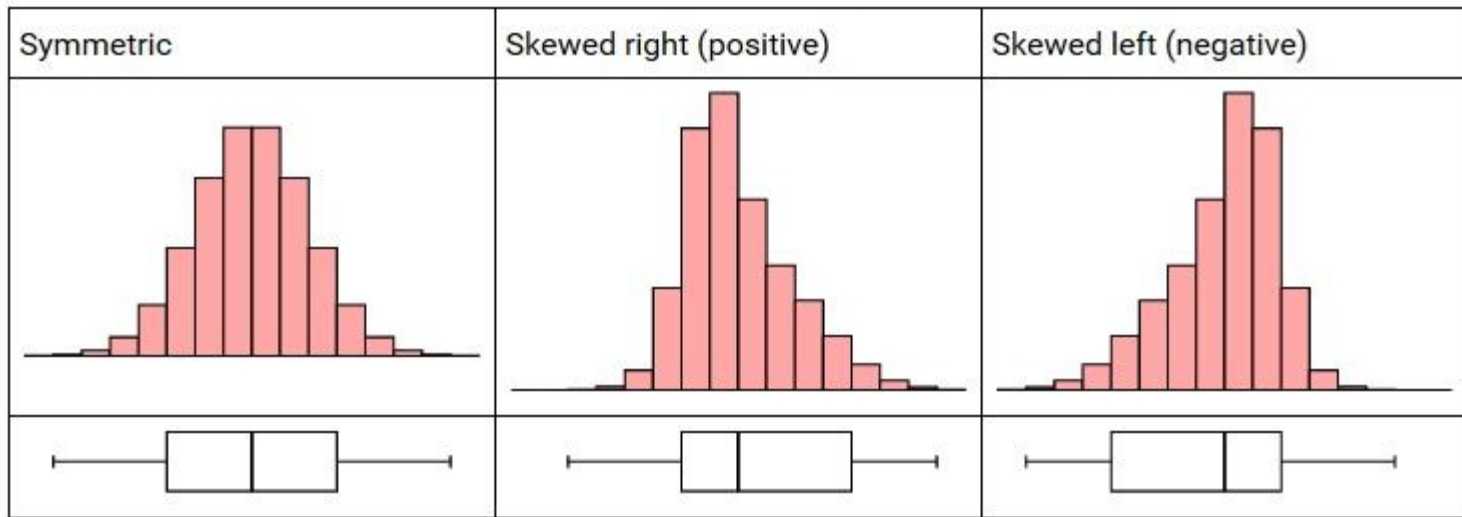


Constructing Boxplots (cont.)

4. Add the **outliers** by displaying any data values beyond the fences with special symbols.
- We often use a different symbol for “far outliers” that are farther than 3 IQRs from the quartiles.



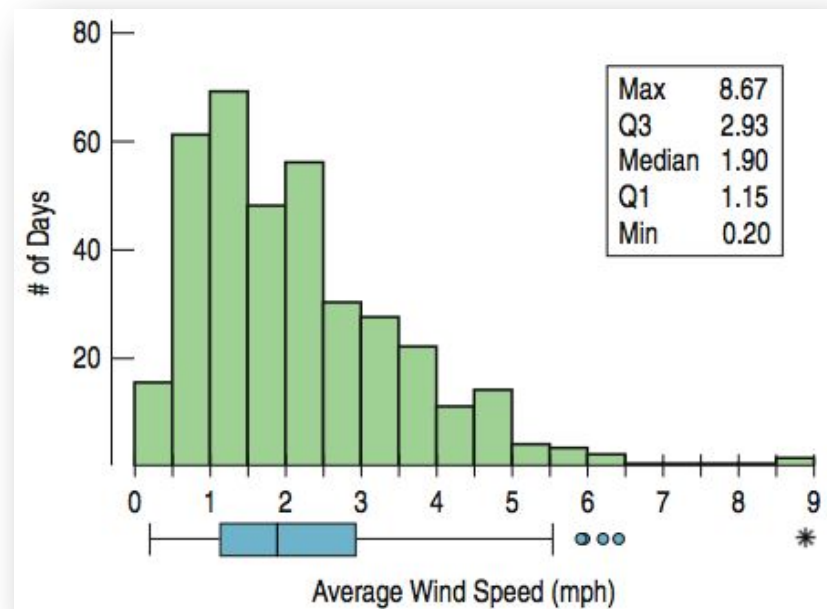
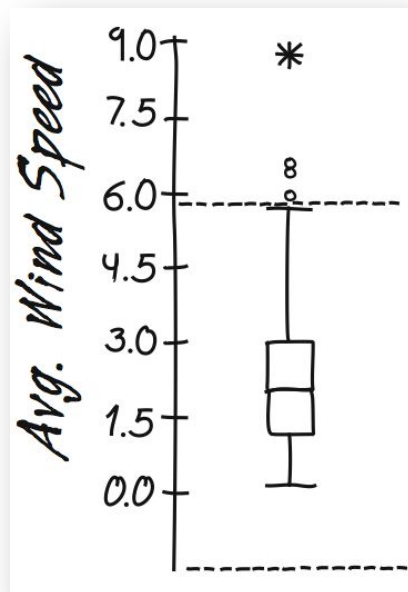
Skewness



Classwork

- [soodk6/cpsc375-classwork](#)
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Boxplots vs Histograms



Classwork

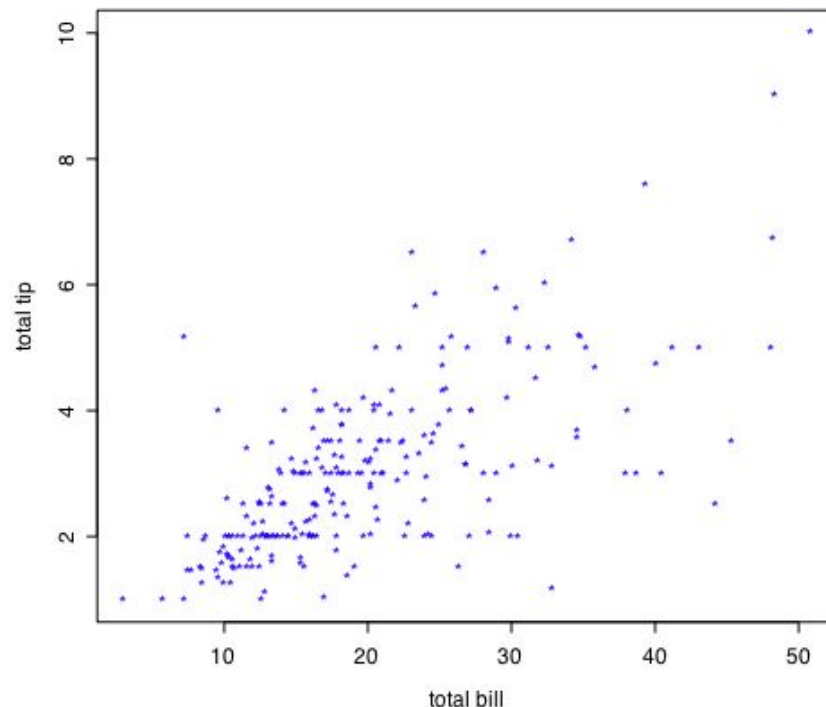
- [soodk6/cpsc375-classwork](#)
–eda.ipynb – Part 3

Pairs of Variables Visualization

- It is almost always more interesting to compare groups
- Two numerical variables:
 - Scatterplot
- One numerical variable, one categorical variable
 - Box plot
- Two categorical variables
 - Bar graph

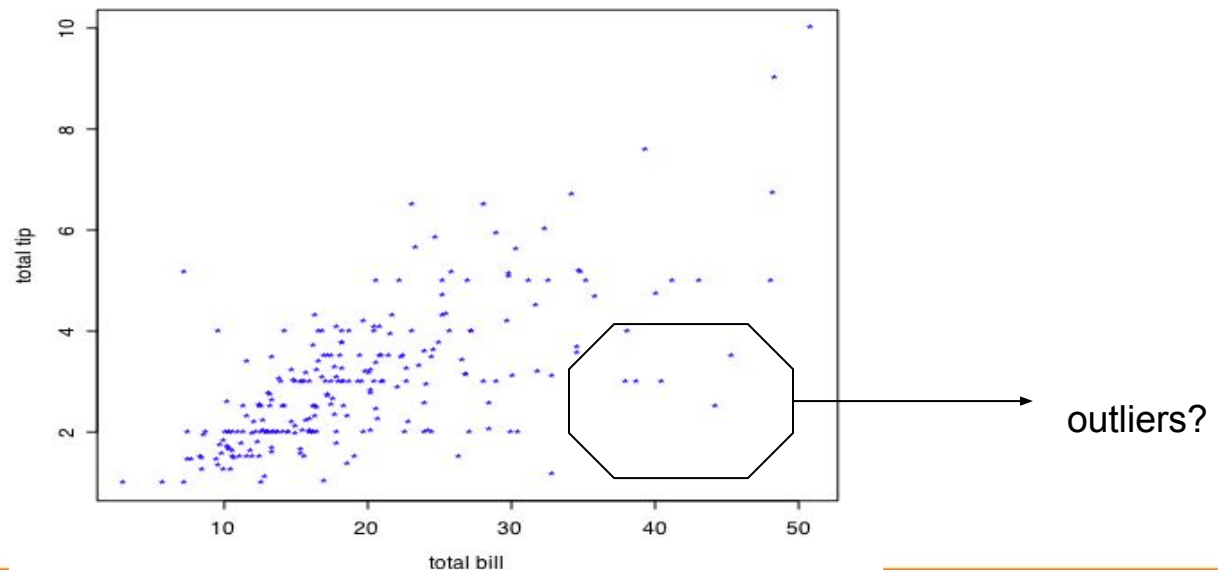
Two Numerical Variables

- For two numeric variables, the scatterplot is the obvious choice



Scatterplots

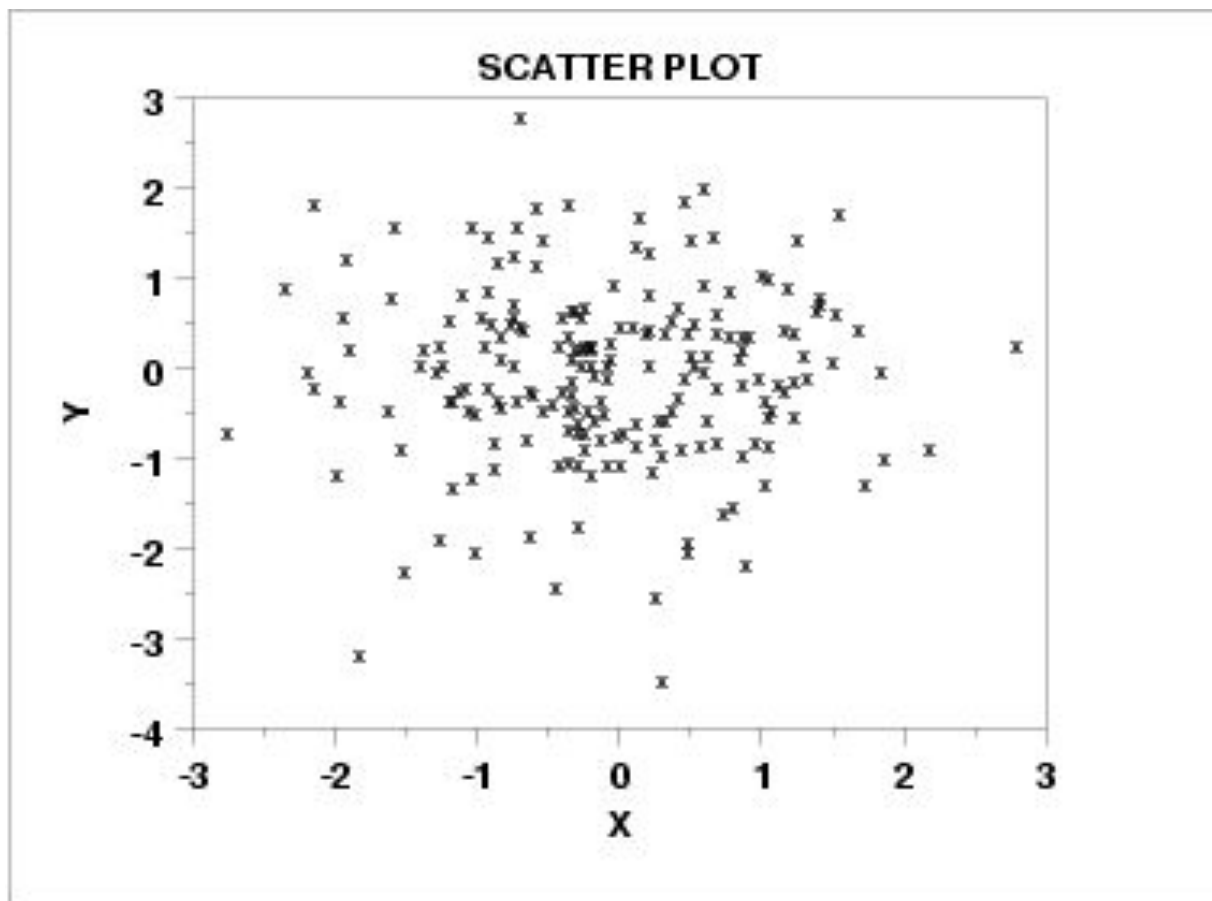
- standard tool to display relation between 2 variables
 - e.g. y-axis = response, x-axis = suspected indicator
- Questions to answer:
 - x,y related?
 - linear
 - quadratic
 - other
 - variance(y) depend on x?
 - outliers present?



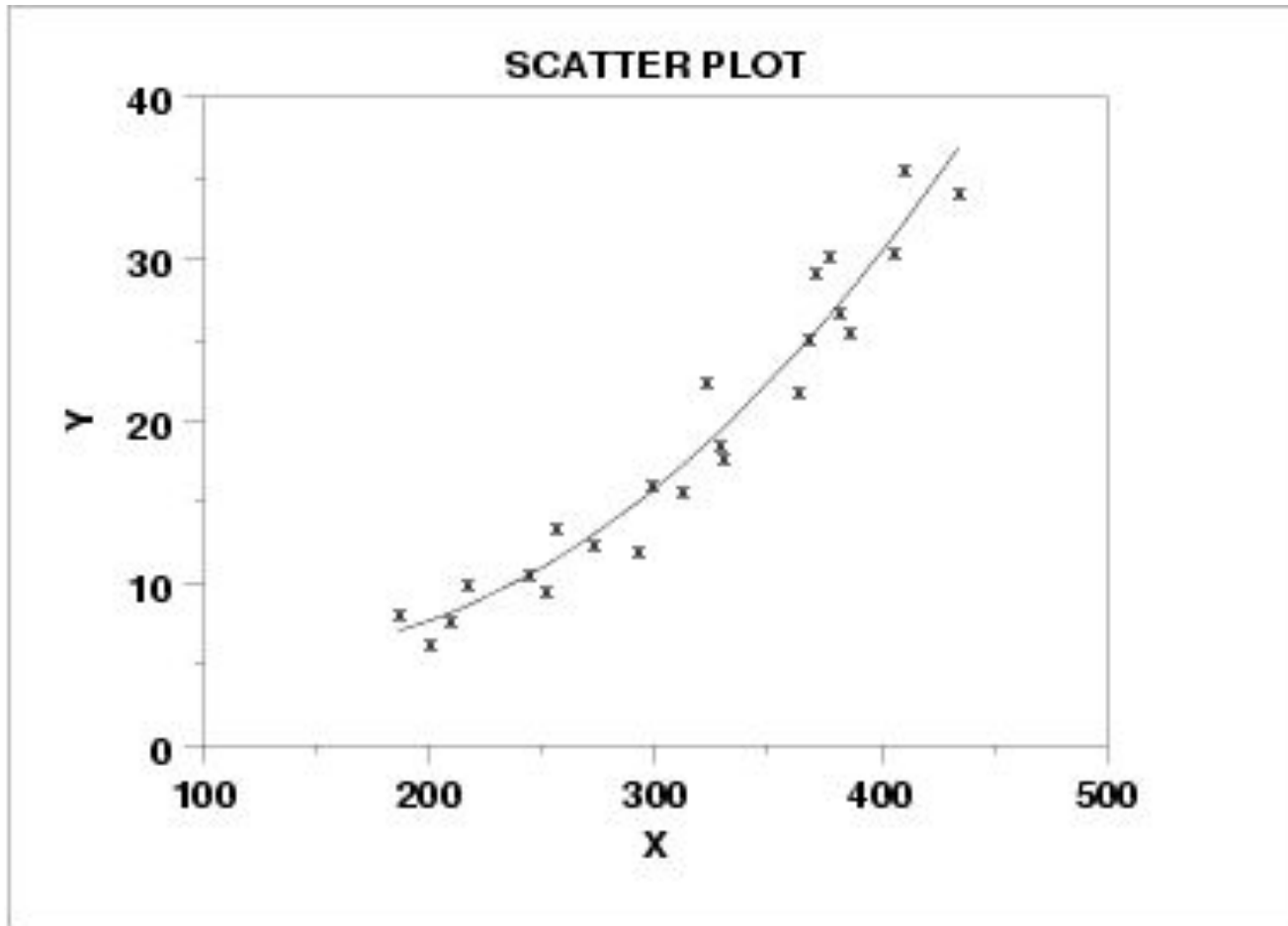
Scatterplots of iris variables

- Is there a relationship between variables?

Scatter Plot: No apparent relationship



Scatter Plot: Quadratic relationship



One numerical variable, one categorical variable

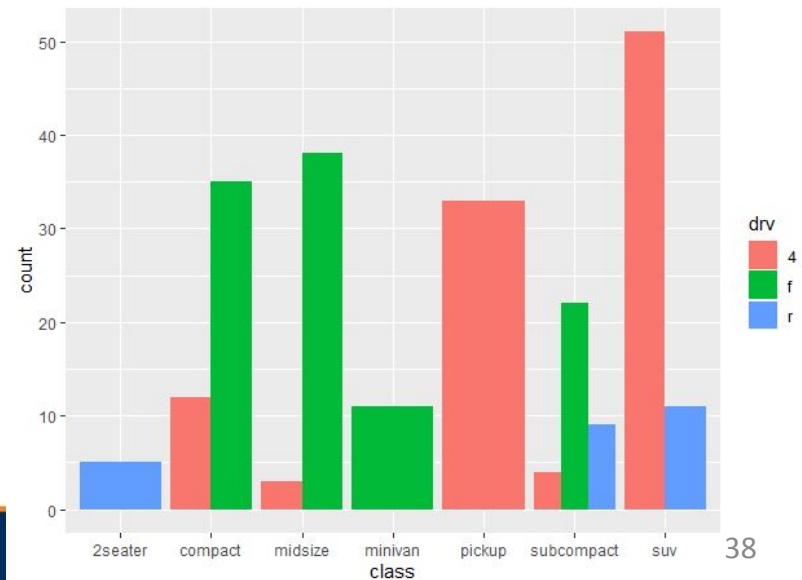
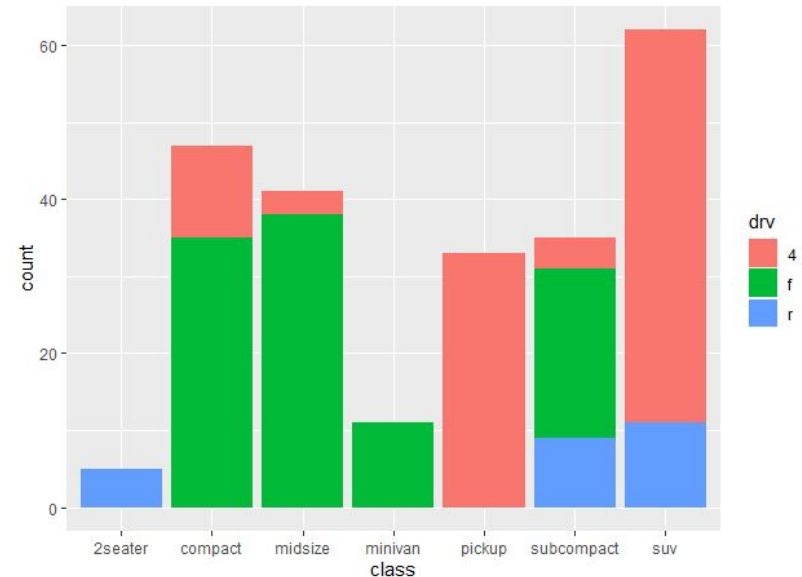
- `ggplot(data=iris) +
 geom_boxplot(mapping=aes(y='Sepal.Width',
x='Species'))`
- **Categorical variable**
 - “Factor” in R
 - Takes one of a small set of values
- **Box plot**
 - x-axis is categorical variable
 - y-axis is numerical

Two categorical variables

- Stacked/dodged bar graph

```
> ggplot(data=mpg) +  
  geom_bar(mapping =  
    aes(x='class',  
        fill='drv'))
```

```
> ggplot(data=mpg) +  
  geom_bar(mapping =  
    aes(x='class',  
        fill='drv'),  
  position = "dodge")
```

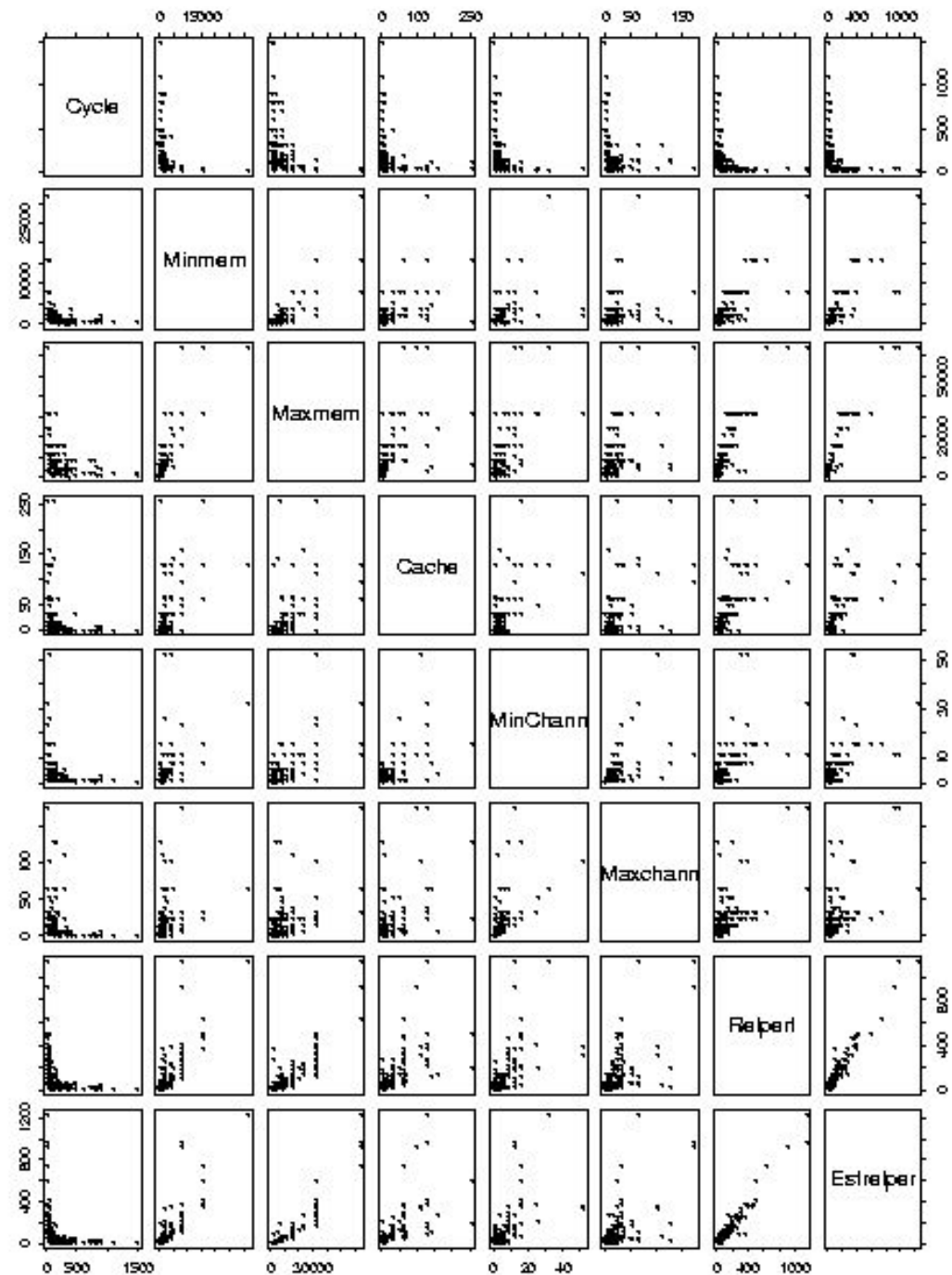


Classwork

- [soodk6/cpsc375-classwork](#)
–eda.ipynb – Part 4

More than two variables

Pairwise scatterplots



Review

- Write in two minutes
 - List the most important things you learned today
 - What question remains in your mind

Summary Statistics

- Summarize numerical variable values
 - mean, median, ...
- mean() method in Pandas, function in NumPy
 - Axis:
 - 0: rows (“index”)
 - 1: columns
 - Methods:
 - mean
 - median

Generate summary statistics of iris variables

- `iris.mean()` # doesn't work
- `np.mean(iris[:,0:4])` # doesn't work
- `iris.iloc[:,0:4].mean()`
- `iris.mean(numeric_only=True)`
- `np.mean(iris.iloc[:,0:4], axis=0)`

Mentimeter



Announcements

- HW3
 - Due soon